Project Design Phase-II
Data Flow Diagram & User Stories

Date	17 October 2026
Project Name	Metro Ticket Generating System using ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams

A Data Flow Diagram (DFD) shows how data enters, is processed by, and leaves a system, including where records are stored. In this project, a passenger submits journey details through the Service Catalog, fare calculation is performed, Flow Designer automates ticket creation, and the system returns a QR-code digital ticket and confirmation.

System flow: Passenger selects source and destination and enters travel details → Service Catalog validates the request → fare is calculated → Flow Designer creates the booking/ticket record → QR-code ticket is generated → Notifications confirm the booking.

Passenger	**1. Service Catalog Booking Form**	**2. Validate Input Calculate Fare**	**3. Flow Designer Create Ticket**
Journey details →	Validated booking data →	Fare + booking details →	Ticket creation
	D1: Booking / Ticket Record	**Digital Ticket + QR Code**	**Notifications**

Processing Steps

1. Passenger opens the ServiceNow portal and selects the Metro Ticket Booking item in the Service Catalog.
2. Passenger enters source station, destination station, and required travel details.
3. The form validates required fields and station selections.
4. The system automatically calculates the fare using configured fare rules.
5. Flow Designer automates creation of the booking/ticket record.
6. A digital ticket with QR code is generated and Notifications send booking confirmation.

User Stories

Use the table below to list user stories for the Metro Ticket Generating System. Priorities and releases are suggested and can be adjusted to your implementation.

User Type	Functional Requirement (Epic)	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Passenger	Ticket Booking	MTS-1	As a passenger, I can open the Metro Ticket Booking item in the Service Catalog to start booking.	Catalog item opens the booking form.	High	Sprint-1
Passenger	Journey Details	MTS-2	As a passenger, I can select source/destination stations and enter travel details.	Required fields are validated; source and destination cannot be identical.	High	Sprint-1
Passenger	Fare Calculation	MTS-3	As a passenger, I can see the fare calculated automatically for my journey.	Fare is calculated using configured rules and shown before submission.	High	Sprint-1
Passenger	QR Ticket Generation	MTS-4	As a passenger, I receive a digital ticket with a QR code after a valid booking.	Ticket reference and QR ticket are generated with journey details.	High	Sprint-1
Passenger	Booking Confirmation	MTS-5	As a passenger, I receive a notification after the ticket is generated.	Notification confirms successful ticket creation and provides ticket details/access instructions.	High	Sprint-1
Passenger	Ticket Access	MTS-6	As a passenger, I can access my digital ticket when needed for travel.	Generated ticket can be viewed through the supported portal/reference.	Medium	Sprint-2

User Stories (continued)

Additional user stories for administration, automation, and future enhancements.

User Type	Functional Requirement (Epic)	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
System / Admin	Booking Records	MTS-7	As an administrator, I can view booking records to support ticket tracking.	Successful booking stores journey details, fare, ticket reference, and time.	High	Sprint-1
System / Admin	Flow Automation	MTS-8	As an administrator, I want Flow Designer to automate booking and ticket creation.	Valid submission triggers the flow and creates the ticket record.	High	Sprint-1
System	Validation & Errors	MTS-9	As a system, I validate booking data and show clear messages for invalid requests.	Invalid or incomplete requests cannot generate a ticket and show an error.	High	Sprint-1
Project Team	Future Enhancements	MTS-10	As a project team, we may add cancellation, refund handling, or QR validation later.	Optional enhancements are planned separately from the core booking workflow.	Low	Sprint-2

Scope note: Service Catalog, Flow Designer, Notifications, automatic fare calculation, and QR-code ticket generation follow the supplied project description. Ticket retrieval, tracking, cancellation, refunds, and advanced QR validation are suggested features; keep them only if they match your implementation.